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LAB 5 — CROSS-REFERENCE AND SURF A DRAWING SET

For

Technical Drawing with AutoCAD Electrical

TGS Ref No: TGS-2023037466

Conducted by

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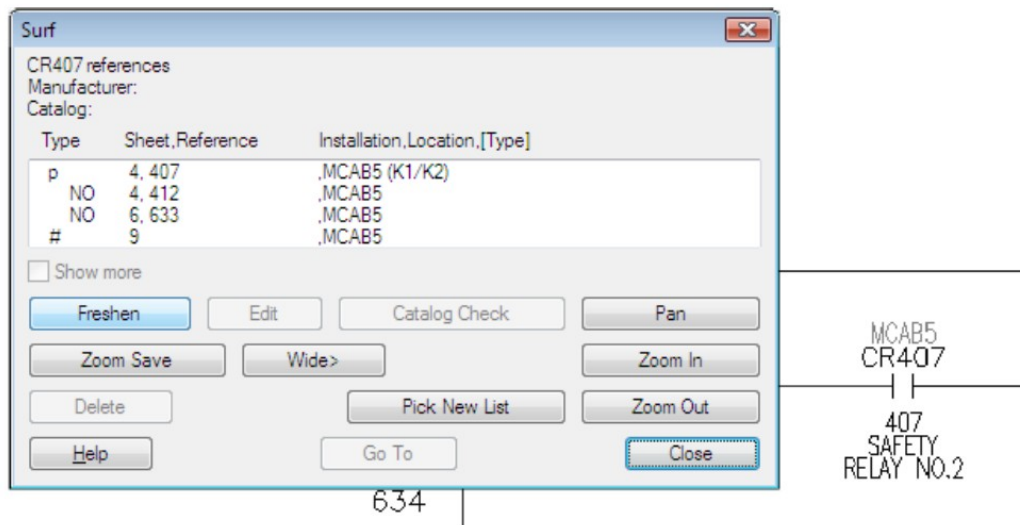
Lab 5 — Cross-Reference and Surf a Drawing Set

Topic 02: Analysing Technical Drawings

Maps to: A3, A4 — review a multi-sheet set for accuracy with cross-references, Surfer and signal arrows

Goal

Link a parent coil to its child contacts, break a wire across sheets with source/destination signal arrows, then audit the set with Surfer and Mark/Verify — how a checker reviews a real drawing set.



What you'll build

A cross-referenced two-sheet circuit you can navigate with Surfer, plus a Mark/Verify change report. (Tools: Cross-referencing · Signal Arrows · Surfer · Mark/Verify · Reference DWG.)

Starter files (in this folder)

- 10-1_Working_with_References.dwg — source: github.com/PacktPublishing/AutoCAD-2025-Best-Practices-Tips-and-Techniques
- 10-2_HVAC_Plan.dwg — source: github.com/PacktPublishing/AutoCAD-2025-Best-Practices-Tips-and-Techniques

Step-by-step

1. Insert a relay coil (parent) on sheet 1 and a NO contact (child) on sheet 2 of your project; assign the child to the parent in the Insert/Edit Child dialog.

AECOMPONENT → coil / contact

2. Update cross-references and read the parent's contact grid — it reports where every child lives.

AEXREF

3. Break a wire network across the two sheets with a Source arrow (assign a code) and a matching Destination arrow.

AESOURCE · AEDEST

4. Right-click the coil and Surf: jump between the parent, children and signal arrows from the Surf list.

AESURF

5. Mark the project (Mark/Verify), move one component, then run Verify to get the change report.

AEMARKVERIFY

6. Open the supplied reference drawings and identify which sheets are reference-only in Project Manager.

OPEN 10-1_Working_with_References.dwg

Test it

Surfer jumps coil → contact → signal arrow across sheets without manual searching, and the Verify report lists exactly the one component you moved — the set is auditable for accuracy.